

AI-Based Automated CERT and Cybercrime Investigation System

Final Year Project (FYP) Defense Proposal

Institution	KICSIT, Institute of Space Technology (IST) Campus, Kahuta, Rawalpindi
Department	Computer Science
Degree / Batch	BSCS, 2023-2027
Area of specialization	Cyber Security (Digital Forensics)
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Deliverable type	Software system (academic prototype)
Duration	12 months

1. Domain Introduction

Cybersecurity incident response, Computer Emergency Response Teams (CERT), and digital forensics workflow automation.

2. Problem Statement

Manual cybercrime handling causes delays, risks evidence contamination, and overburdens investigative agencies. Victims get no immediate guidance, complaints are unevenly triaged, evidence handling is hard to trace, and incident response is separate from investigation. A structured, AI-assisted workflow under human supervision is needed to preserve evidence integrity and speed up case processing.

3. Project Summary (under 200 words)

The proposed system is a web-based cybercrime complaint, incident-response and investigation platform that covers the lifecycle from public complaint to resolution. It provides complaint registration and tracking, a 24/7 AI first-aid chatbot for safe guidance and preliminary triage, role-based access control, a CERT-oriented incident-response workflow, digital evidence management with SHA-256 hashing, chain of custody and audit logs, and integration of selected open-source forensic tools (Autopsy, Wireshark, ExifTool). AI-assisted analysis (entity extraction, timelines, summarization, search) supports investigators. All AI output is decision support only; authorized humans make every important decision. The system is an academic prototype evaluated on synthetic and public data. It manages forensic workflows but does not replace professional forensic tools, prove guilt, or establish legal admissibility.

4. Objectives

1. Develop an integrated platform for cybercrime complaint, case and investigation management.
2. Automate initial incident response through a 24/7 AI chatbot and CERT-oriented workflows.
3. Provide secure role-based access, evidence management, chain of custody and investigation support.
4. Support selected digital-forensics workflows and AI-assisted analysis for investigators.
5. Evaluate integrity mechanisms, including whether blockchain logging adds value over hashing plus audit logs.

5. Definitions

- **Incident:** the alleged cyber event itself.
- **Complaint:** the report submitted by the affected person. It is not automatically a confirmed crime.
- **Case:** a matter formally accepted after screening.

6. Scope

In scope: complaint to closure workflow; two separate AI components; evidence registry, hashing, chain of custody; forensic task and result management with selected tools; CERT-oriented workflow; RBAC, audit logging, secure file handling; security testing; synthetic, public or authorized data only.

Not claimed: crime prediction or GIS mapping; universal forensic acquisition; automatic identification of criminals or guilt; legal admissibility; a national CERT; production government deployment; use of real citizen data.

7. System Design

Lifecycle: Public Complaint -> AI Guidance and Triage -> Screening -> Case Creation -> Assignment -> Investigation -> Evidence Management -> Forensic Examination -> CERT Workflow (where applicable) -> AI-Assisted Analysis -> Review and Reporting -> Closure

7.1 Modules

Module	Main functions
Public complaint	Registration, category, description, evidence upload, reference number, status view
AI first-aid chatbot (AI-A)	Safe guidance, evidence-preservation tips, category and urgency suggestion, escalation to human
Screening and case management	Review queue, request info, accept/reject/refer, assignment, tasks, notes
RBAC	Roles, least privilege, MFA for staff
Evidence management	Metadata, SHA-256, storage, access history, case linking
Chain of custody	Events for submit, receive, access, transfer, status change, approval
Forensic workflow	Forensic tasks, tool outputs, findings, examiner notes
CERT workflow	Incident intake, severity, assignment, response tracking, evidence preservation, link to case
AI investigation support (AI-B)	Classification, entity extraction, timeline, correlation, summaries, search
Reporting and closure	Findings, review and approval, reports
Integrity and audit	Hash verification, append-only audit log, optional blockchain log
Security	Secure upload, validation, encryption, backup

7.2 Two separate AI components

- **AI-A (victim-facing):** first-aid guidance, triage, escalation. Built with retrieval over a reviewed guidance knowledge base and fixed safety rules. It never declares guilt, gives legal decisions, claims certainty about an attack, or counts as evidence.
- **AI-B (officer-facing):** assists investigators with extraction, timelines, correlation and summaries. Output is labeled *AI-generated*, *requires human review*.

7.3 Forensic integration

Evidence	Tool	Role
Disk image (sample/public)	Autopsy	Examiner uses tool; platform stores output, links evidence, records custody
Network capture (PCAP)	Wireshark / tshark	Import selected results

Evidence	Tool	Role
Images and documents	ExifTool	Automated metadata extraction
Chats, emails, logs	Custom parsers	Text extraction for AI-B

Mobile and CCTV work uses pre-acquired or synthetic artifacts only.

7.4 Evidence integrity

SHA-256 on upload with on-demand re-verification, plus hash-chained append-only audit logs. Hashing detects change; it does not prove authenticity. Blockchain is an optional experiment that stores only hashes and selected custody events on a private test chain.

Research question: Does blockchain-based evidence logging provide meaningful benefit over a secure centralized database with hashing and audit logs in this workflow? If not, the report will state that result.

7.5 RBAC roles

Public complainant, screening officer, investigation officer, evidence officer, forensic examiner, CERT officer, supervisor/reviewer, administrator. The administrator manages users and configuration but has no automatic access to evidence.

8. Project Implementation Method

Iterative (agile) development: requirements analysis, system design, database development, implementation of complaint, CERT, investigation and evidence modules, AI integration, security controls, forensic tool integration, then testing with controlled and synthetic cybercrime scenarios. Each phase is reviewed with the supervisor.

Proposed tools (free/open-source): React or Next.js, FastAPI or Django, PostgreSQL, MinIO or encrypted local storage, Python NLP (spaCy, scikit-learn, sentence-transformers), open-source LLM or API for retrieval and summaries, Hardhat/Ganache (experiment), Docker Compose, OWASP ZAP. Final choices confirmed with the supervisor.

Data: synthetic complaint scenarios, public phishing/fraud text datasets (license permitting), public sample forensic images and PCAPs, self-generated screenshots, chat exports and logs. No real personal data.

9. Key Milestones

#	Elapsed time	Milestone	Deliverable
1	Months 1-2	Literature review, existing-system analysis, requirements, tool testing	Requirements document, literature review, workflow design
2	Months 3-5	Core platform: complaints, RBAC, case workflow, evidence, hashing, custody, audit log	Working core system
3	Months 6-7	AI chatbot and CERT workflow	Chatbot module, CERT module
4	Months 8-9	AI investigation analysis and forensic tool integration	AI-B module, forensic workflow
5	Month 10	Blockchain experiment and reporting module	Experiment results, report module
6	Month 11	Full evaluation and security testing	Evaluation and security reports
7	Month 12	Documentation, thesis, demo, defense	Final thesis, demo, presentation

Team split (suggested): backend, RBAC, evidence and security; AI-A, AI-B and evaluation; frontend, forensic integration, CERT workflow and blockchain experiment. All members share testing and documentation.

10. Final Deliverable

Software system (tick as on form). A web-based platform with public complaint submission and case tracking, 24/7 AI first-aid chatbot, RBAC, CERT-oriented incident response, digital evidence management, chain of custody, hashing and audit logs, selected open-source forensic workflows, and AI-assisted investigation, with human oversight for all decisions.

11. Equipment Required (estimated)

Item	Purpose	Estimated cost
Existing laptops/PCs of group members	Development and testing	Rs. 0 (own)
Open-source software (all tools listed above)	Development, forensics, testing	Rs. 0
Cloud VPS or hosting for demo (optional)	Deployment and demo	approx. Rs. 10,000
LLM API credits (optional, if no local model)	Chatbot and summaries	approx. Rs. 10,000
External storage drive	Test datasets and backups	approx. Rs. 8,000
Total (approximate)		approx. Rs. 28,000

Costs are estimates and should be confirmed before submission.

12. Evaluation Plan

Area	Method	Metrics
Complaint module	Functional and validation tests	Success rate, validation errors caught
RBAC	Role-permission test matrix	% unauthorized actions blocked
Evidence and hashing	Tamper test	Detection rate, false alarms
Chain of custody	Scenario replay	Completeness, traceability
Blockchain vs centralized	Tamper and performance test	Detection, overhead, complexity
AI-A chatbot	Scenario test set, rubric review	Category accuracy, relevance, safety violations, escalation correctness
AI-B analysis	Labeled samples	Precision, recall, F1, summary quality by human review
Forensic integration	Known test artifacts	Correct import and extraction
Security	OWASP Top 10 tests	Vulnerabilities found and fixed
Performance and usability	Load test, user survey	Response time, SUS score

13. Related Work and Novelty Position

Complaint portals, incident-response case platforms (e.g., TheHive), forensic suites (e.g., Autopsy), threat-sharing platforms (e.g., MISP) and commercial evidence-management products already exist. This project does not claim that no similar system exists. Its contribution is a focused, evaluated integration of complaint handling, AI first-aid, investigation, evidence traceability, forensic workflow and CERT-style response in one open-source prototype, plus an empirical comparison of blockchain and centralized integrity logging. A full literature review with cited references will be completed in Milestone 1.

14. Benefits

Direct: cybercrime complainants, investigators, CERT/incident-response personnel and authorized officers, through faster complaint handling, structured response, traceable evidence and better coordination.

Indirect: organizations and the general public, through improved awareness, reporting and structured handling of cyber incidents.

15. Risks and Ethics

Risk	Mitigation
Scope too large	Build core modules first; drop optional items if late
LLM hallucination	Reviewed knowledge base, safety rules, human review labels
No real data	Synthetic and public data, limits documented
Forensic tool complexity	Integrate outputs, do not re-implement tools
Blockchain adds no value	Report the negative result honestly
Platform security flaws	Early security testing, least privilege

Ethics: prototype only, no real citizen data, AI output is not proof, the system records custody but does not declare admissibility, and uploaded files are never treated as verified evidence automatically.

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